

The empirical recurrence for OEIS A183920, proved from an exact model

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Abstract

OEIS A183920 carries an empirical recurrence of order 11. It is true, and it is decidable rather than empirical. The entry's sequence is given exactly by an Ehrhart quasi-polynomial of the multiset cone, from which a provably satisfies a MONIC linear recurrence of order $S = 13$, derived below and not assumed. The residual of the conjectured recurrence satisfies the same one, so whether it annihilates a is settled by a finite exact computation. No transfer matrix is involved and the count is not a walk.

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1 The conjecture

OEIS A183920 is “Number of nondecreasing arrangements of n numbers in $-2..2$ with sum zero and sum of squares not greater than $n*6/3$.”. It has offset 1 and begins

$$1, 2, 4, 6, 9, 13, 17, 24, \dots$$

The entry states, as a conjecture contributed by its author and never marked settled:

$$\text{Empirical: } a(n) = a(n-1) + 2*a(n-3) - a(n-4) - a(n-5) - a(n-6) - a(n-7) + 2*a(n-8) + a(n-10) - a(n-11).$$

As of the “Last modified” line on the live entry (Jun 23 2026 13:34:22, revision 13) this is still recorded as empirical, and nothing on the entry records it as proved.

2 The count is an Ehrhart quasi-polynomial

A nondecreasing arrangement of n numbers from $-2..2$ is a MULTISSET, so it is its counts c_v for $v = -2, \dots, 2$, and every condition the entry names is linear in those and in n :

$$\sum_v c_v = n, \quad \sum_v v c_v = 0, \quad c_v \geq 0 \text{ and}$$

$$\sum_v v^2 c_v \leq 2n.$$

Nothing here is a walk: the LENGTH grows and the alphabet is fixed, so no window counts it. Use the first equation to eliminate n ; every remaining condition is then homogeneous in c alone, the admissible c form a finite union of relatively open rational cones in $\mathbf{R}^{2+1}$, and $a(n)$ counts their lattice points at height $\sum_v c_v = n$. It is therefore an Ehrhart quasi-polynomial, and the count itself is an exact integer dynamic programme over v carrying the running sums.

3 The bound

The period is read off the arrangement rather than assumed. Every ray of every cell is cut out by as many independent hyperplanes as there are variables less one, so by Cramer its primitive generator is the vector of cofactors, and its height $\sum_v c_v$ is what the period divides. A ray leaving the region — some c_v negative, or the wrong side of one of the two other conditions — bounds no cell and is dropped. Writing T for the heights that survive, the annihilator is $\prod_{d|t \in T} \Phi_d(z)^{m_d}$ with m_d at most the dimension of the cone, which the equality $\sum_v v c_v = 0$ drops to 4; one further factor z covers the origin, the one cell with no ray, whose series is the constant 1. That gives a monic annihilator of order $S = 13$, which was then asked for terms the model had not supplied and reproduced them.

4 The proof

Lemma 1. *Let $A(z) = z^S - \sum_{j < S} \alpha_j z^j$ be the monic annihilator derived above and $q(z) = z^{11} - \sum_i c_i z^{11-i}$ the characteristic polynomial of the conjectured recurrence. The residual $r(n) = a(n) - \sum_i c_i a(n-i)$ is a linear combination of shifts of a , so A annihilates r as well. Since $A(0) \neq 0$ the recurrence it gives runs in both directions, and S consecutive zeros of r force r to vanish at every later index.*

Proof. A annihilates a , hence every shift of a , hence any linear combination of shifts; that is r . A monic recurrence with nonzero constant term determines each term from the S before it and each from the S after it, so a run of S zeros propagates. $\square$

Theorem 2.

$$a(n) = a(n-1) + 2a(n-3) - a(n-4) - a(n-5) - a(n-6) - a(n-7) + 2a(n-8) + a(n-10) - a(n-11)$$

for every $n > 0$.

Proof. The terms $a(0), \dots$ were computed exactly in integer arithmetic from the model, extended by A where the model itself was not evaluated, and $r(n)$ formed for each index. Those integers vanish from $n = 0 + 1$ onwards, and the run exceeds $S + 11$, which by Lemma 1 settles every larger index at once. $\square$

5 Verification

Three checks, each independent of the algebra above.

First, the model reproduces all 53 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry: it is built by reading the entry's English, and a misreading gives different counts.

Second, the conjectured recurrence was evaluated directly on those published terms, with no model involved, and holds wherever the proved range and the data overlap.

Third, the derived annihilator A was itself tested on terms it had not been given: the model was evaluated beyond the S values that determine it and A reproduced them. A bound that is too small fails this test, which is why it is run.

References

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